MIL-P-46412C (MU)
29 JULY 1966
SUPERSEDING
MIL-P-46412B (MU)
25 JANUARY 1963

MILITARY SPECIFICATION

PROJECTILE, 155MM, GAS, M121A1 ASSEMBLING, MARKING AND PACKING

1. SCOPE

1.1 This specification covers the assembling, marking, and packing for one type of projectile designated as Projectile, 155MM, Gas, Persistent, VX, M121A1, or designated as Projectile, 155MM, Gas, Non-Persistent, GB, M121A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons; General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

MIL-STD-109 — Quality Assurance Terms and Definitions

MIL-STD-1168 — Lot Numbering of Ammunition

MIL-STD-1235 — Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

XP-112889 — Projectile, 155MM, Gas, Persistent, VX, M121E1, Simulant Filled.

7549275 — Packing and Marking of Pallet for 155MM Projectiles, Empty or Loaded.

8861029 — Projectile, 155MM, Gas, M121A1, Loading Assembly.

FSC 1320

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PUBLICATIONS

ARMY

ET–8861029 — Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. The assembly shall comply with all requirements specified on Drawing (dwg.)8861029, and with all requirements specified in applicable specifications.

3.3 Proving ground.

3.3.1 *Premature burst.* There shall be no functioning of the projectile in the gun bore or in flight before impact.

3.3.2 *Functioning.* The projectile shall function with sufficient fragmentation to form a visible cloud.

3.4 Workmanship. The projectile shall be dry and free of chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(a) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority will not be required with subsequent

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certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL–STD–105 and shall be numbered in accordance with MIL–STD–1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Each lot of projectiles shall contain:

(a) VX or GB loaded filling assemblies from one manufacturer in accordance with the same drawing, same drawing revision, same specification and specification revision of one loading lot interfix number.

(b) Projectile bursters from not more than one interfix lot.

(c) Supplementary charges from not more than one interfix lot.

(d) Burster projectile support assemblies from not more than one interfix lot.

4.2.2 *Examination.* Inspection for Critical defects (and Major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with Standard MIL–STD–105 except that continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Projectile burster, prior to assembly to projectile (see dwg. 8861029).

Categories	*Defects*	*Method of Inspection*	*Code No. (see 6.2)*
Critical:			
1.	Cushioning pad missing	Visual	01001
2.	Closure missing	Visual	01002
Major:	None defined.		
Minor:	None defined.		

4.2.2.2 Loading assembly, prior to assembly of support assembly (see dwg. 8861029).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:			
1.	Projectile burster inverted	Visual	02001
Major:	AQL 0.40 percent		
101.	Projectile burster missing	Visual	02002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	02003

4.2.2.3 Projectile burster support assembly, prior to assembly to projectile (see dwg. 8861029).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:			
1.	Cushioning pad missing from bottom of burster	Visual	03001
Major:	None defined.		
Minor:	None defined.		

4.2.2.4 Fuze adapter, prior to assembly to projectile (see dwg. 8861029).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Coating on fuze adapter threads missing	Visual	04001
102.	O-ring missing or damaged	Visual	04002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	04003

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4.2.2.5 Loading assembly, prior to assembly of fuze adapter (see dwg. 8861029).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Burster projectile support assembly missing	Visual	05001
102.	Supplementary charge missing (when applicable at this stage of assembly)	Visual	05002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	05003

4.2.2.6 Loading assembly, prior to assembly to lifting plug (see dwg. 8861029).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:			
1.	Shell zone weight marking incorrect by two or more zones	Approved Scale	06001
Major:	AQL 1.00 percent		
101.	Shell zone weight marking incorrect by one zone	Approved Scale	06002
102.	Length, max	Gage	06003
103.	Fuze adapter improperly staked	Visual	06004
104.	Spacer missing	Visual	06005
105.	Supplementary charge missing (when applicable at this stage of assembly)	Visual	06006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	06007

4.2.2.7 Lifting plug, prior to assembly to projectile (see dwg. 8861029).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Corrosive preventive compound missing from threads	Visual	07001
102.	Gasket missing or damaged	Visual	07002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	07003

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4.2.2.8 Loading assembly (see dwg. 8861029).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:			
1.	Marking misleading or unidentifiable, including incorrect color code	Visual	08001
2.	Projectile with incorrect type of gas (not as required for the particular lot being loaded)	Visual	08002
Major:	AQL 0.65 percent		
101.	Diameter of bourrelet (if shell is retouched)	Gage	08003
102.	Torque test failure of lifting plug	Test	08004
103.	Grommet missing	Visual-Manual	08005
104.	Any weight zone punch mark missing or inadequate	Visual	08006
Minor:	AQL 1.5 percent		
201.	Total length	Gage	08007
202.	Grommet obviously loose	Visual-Manual	08008
203.	Protective coating damaged (total bare area in excess of ¼ square inch)	Visual	08009
204.	Evidence of poor workmanship (see 3.4)	Visual	08010

4.2.2.9 Pallet, prior to strapping (see dwg. 7549275).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Cover assembly missing	Visual	09001
102.	Unauthorized mixing of lots on same pallet	Visual	09002
Minor:	AQL 1.0 percent		
201.	Sleeve missing	Visual	09003
202.	Incorrect number of projectiles	Visual	09004

4.2.2.10 Pallet (see dwg. 7549275).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Board broken	Visual	10001
102.	DOD symbol missing or incorrect	Visual	10002

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Categories	*Defects*	*Method of Inspection*	*Code No.*
103.	Strapping missing or loose	Visual	10003
Minor:	AQL 1.00 percent		
201.	Marking misleading or unidentifiable	Visual	10004
202.	Strapping improperly engaged	Visual-Manual	10005

4.2.3 *Testing.*

4.2.3.1 *Premature burst.* (see 3.3.1)—Critical defect—Code No. 08011.

4.2.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirement specified, twenty-five sample projectiles shall be selected at random from each lot for this test. If any projectile fails to comply with the requirements of 3.3.1, the lot shall be rejected.

4.2.3.1.2 After the three consecutive lots have met the criteria of 4.2.3.1.1, ten sample projectiles shall be selected at random from each lot for this test. If any projectile fails to comply with the requirements of 3.3.1, the lot shall be rejected.

4.2.3.2 *Functioning.* (see 3.3.2)—Major defect—Code No. 08012.

4.2.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirement specified, the sample cartridges fired in the test for premature burst (see 4.2.3.1.1) shall be observed for functioning. If more than two projectiles fail to comply with the requirements of 3.3.2, the lot shall be rejected.

4.2.3.2.2 After the three consecutive lots have met the criteria of 4.2.3.2.1, the sample cartridges fired in the test for premature burst (see 4.2.3.1.2) shall be observed for functioning. If more than one projectile fails to comply with the requirements of 3.3.2, the lot shall be rejected.

4.2.4 *Inspection Equipment.* Equipment Tabulation Number ET–8861029 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection equipment.* The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Proving ground test.* The tests shall be conducted at a Government proving

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ground in accordance with the applicable Acceptance Test Procedure. The sample projectile shall be loaded with a simulated filler in accordance with dwg. XP–112889 and shall be fired in a gun for which the projectile is standard and observation made for conformance to all applicable requirements.

4.3.1.1 *Premature burst and functioning.* The shells shall be assembled with M508 point detonating fuzes and fired using propellant adjusted to give 112 percent of the average pressure given by the propellant loaded in the particular lot used. Observation shall be made for premature bursts, indication of metal parts separation in the gun bore or in flight and for sufficient fragmentation to form a visible cloud.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* The projectile shall be preserved and packaged in accordance with dwg. 7549275.

5.2 Packing.

5.2.1 *Level A.* The projectile shall be packed in accordance with dwg. 7549275.

5.3 Marking. Marking of the pallet shall be in accordance with dwg. 754275.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) *Data cards.* Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL–STD–1167.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (SMUPA Form 1010) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the EL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL–D–45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

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CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commerical equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commerical equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35mm microfilm process. Contractors shall submit one copy of the final design as a rolled

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set or flat set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 *Submission of contractor designs.* Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of test data and ammunition data cards. In addition to the normal distribution of records, for loaded items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA–ND

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1320–A110

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